

Two classic problems

The matching problem · The gambler's ruin

STA102 — additional material, with complete proofs

Monsoon Semester 2026 · Ahmedabad University

What is in these slides

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Part A: the matching problem

$\mathfrak{n}$ people collect $\mathfrak{n}$ hats at random. How likely is it that nobody gets their own?

Inclusion–exclusion (with proof), the formula, the limit $1/e$, derangement numbers, a second proof by recurrence, and the distribution of the number of matches.

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Part B: the gambler's ruin

You bet one rupee at a time until you reach a target or go broke. How likely are you to reach the target?

Why the game ends, first-step analysis, the formula (two proofs), rich opponents, bold play, Huygens' 1657 problem, deuce in tennis, and how long the game lasts.

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Prerequisites. The axioms of probability, the addition rule, equally likely outcomes, counting ($n!$, $\binom{n}{k}$), conditional probability, the multiplication rule and the law of total probability. Part B also uses the idea that separate bets are independent — "the coin has no memory". Anything that goes beyond this is flagged when it appears.

Part A: The matching problem

le problème des rencontres

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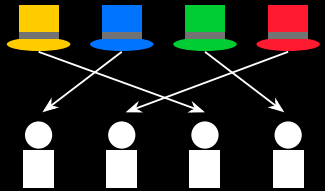
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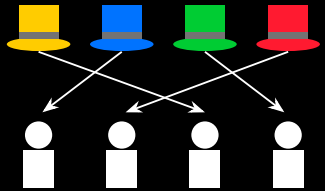
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In every story: what is the probability that nobody gets their own item back? And what is the probability that exactly k people do?

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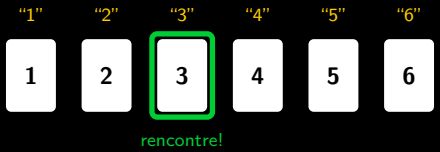
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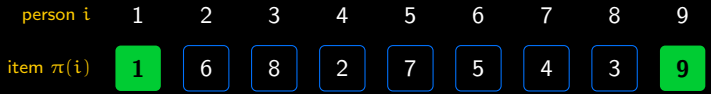
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person i	1	2	3	4	5	6	7	8	9
item $\pi(i)$	1	6	8	2	7	5	4	3	9

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The question becomes: what fraction of the $n!$ permutations have no fixed point?

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Guess: as $n \to \infty$, does P_n go to 0, to 1, or somewhere in between?

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$$S_k = \sum_{|J|=k} P(A_J) = \sum_{i_1 < i_2 < \cdots < i_k} P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}).$$

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We give two proofs: one by induction, valid for any probability; and one by counting how often each outcome is included, valid for finite sample spaces.

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$$P(U) = \sum_{\varnothing \neq I \subseteq \{1..n\}} (-1)^{|I|+1} P(A_I), \qquad P(U \cap A_{n+1}) = \sum_{\varnothing \neq I \subseteq \{1..n\}} (-1)^{|I|+1} P(A_I \cap A_{n+1}).$$

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$J = \{n+1\}$	$+P(A_{n+1})$	$+1 = (-1)^{1+1}$
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So every non-empty $J \subseteq \{1, \dots, n+1\}$ appears exactly once, with sign $(-1)^{|J|+1}$:

$$P\Big(\bigcup_{i=1}^{n+1} A_i\Big) = \sum_{\varnothing \neq J \subseteq \{1..n+1\}} (-1)^{|J|+1} P(A_J) = \sum_{k=1}^{n+1} (-1)^{k+1} S_k.$$

This is the formula for $n+1$ events. ■

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$$\sum_{k=1}^{m} (-1)^{k+1} \binom{m}{k} = 1 - \sum_{k=0}^{m} (-1)^k \binom{m}{k} = 1 - (1-1)^m = 1,$$

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Stopping the alternating sum early gives an upper bound after an odd number of terms and a lower bound after an even number (the Bonferroni inequalities); S_1 alone is the union bound.

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By the multiplication rule (a check): person i_1 gets their own item with probability $\frac{1}{n}$; given that, i_2 gets theirs from the remaining $n-1$ with probability $\frac{1}{n-1}$; and so on:

$$\frac{1}{n} \cdot \frac{1}{n-1} \cdots \frac{1}{n-k+1} = \frac{(n-k)!}{n!}.$$

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The Lemma says every term in S_k is the same, and there are $\binom{n}{k}$ of them:

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Check: $P_3 = 1 - 1 + \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$ and $P_4 = \frac{1}{3} + \frac{1}{24} = \frac{3}{8}$ — matching the lists we made by hand.

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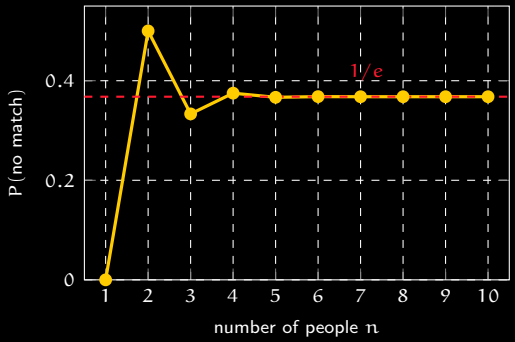
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With $n = 10$ the error is below $1/11! \approx 2.5 \times 10^{-8}$.

Seeing the convergence

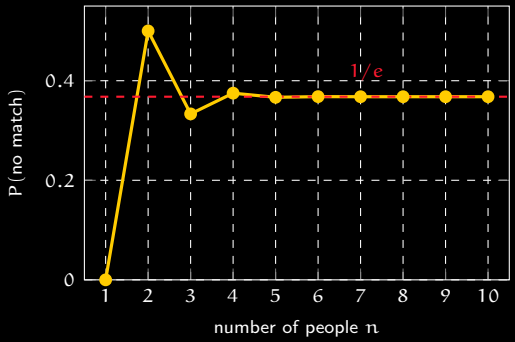
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By $n = 7$ the answer is $1/e$ to four decimal places. A dinner for eight and a stadium of fifty thousand give essentially the same chance — about 36.8% — that nobody gets their own hat.

Derangement numbers

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For example $6!/e \approx 264.87$, and $D_6 = 265$.

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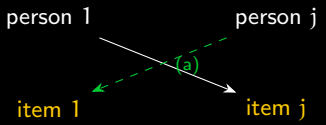
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Check: $D_4 = 3(D_3 + D_2) = 3(2+1) = 9$ and $D_5 = 4(9+2) = 44$.

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Adding up the differences (they telescope), with $P_1 = 0$:

$$P_n = \sum_{k=2}^{n} \frac{(-1)^k}{k!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!},$$

since the $k = 0$ and $k = 1$ terms, 1 and -1, cancel. Same formula, no inclusion–exclusion needed. ■

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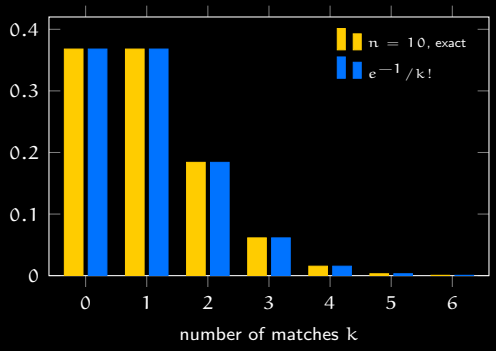
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"Exactly k matches" is the disjoint union of these events over the $\binom{n}{k}$ choices of K, so

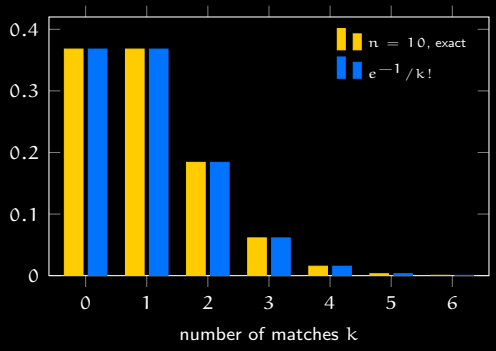
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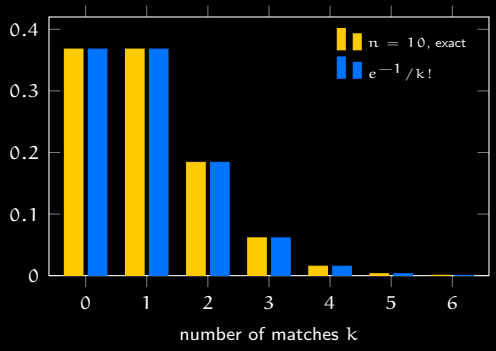


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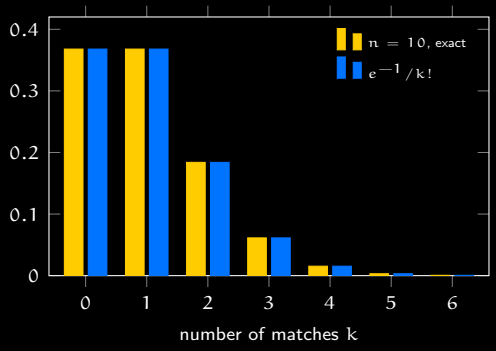
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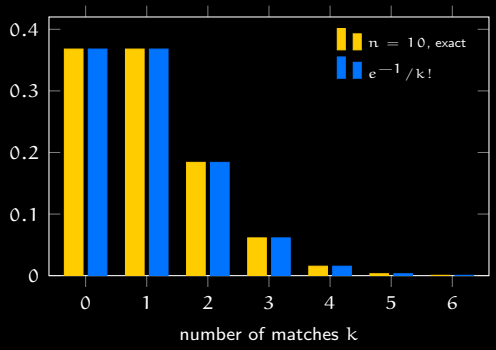


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The numbers $e^{-1}/k!$ form the Poisson distribution with parameter 1. It will appear later in the course as the law of rare events — and here it is, hiding in a cloakroom.

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3. Ten hats thrown in the air (from Blom, Holst and Sandell): the probability that at least half of the ten boys get their own hat back is

$$\sum_{k=5}^{10} \frac{P_{10-k}}{k!} = \frac{829}{226{,}800} \approx 0.0037.$$

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1. Secret Santa among 12 friends: each draws a name at random. The draw works only if nobody draws their own name: probability $P_{12} \approx 0.368$. So about 63% of first attempts have to be redone.
2. A simplified jeu du treize: turn over the 13 cards of one suit, shuffled, counting "ace, two, . . . , king". The dealer wins if at least one card matches its call: probability $1 - P_{13} \approx 0.632$ — a comfortable edge.
3. Ten hats thrown in the air (from Blom, Holst and Sandell): the probability that at least half of the ten boys get their own hat back is

$$\sum_{k=5}^{10} \frac{P_{10-k}}{k!} = \frac{829}{226{,}800} \approx 0.0037.$$

A remark for later in the course: the average number of matches is exactly 1, for every n. You can already check it for $n = 3$: the six permutations $123, 132, 213, 231, 312, 321$ have $3, 1, 1, 0, 0, 1$ matches, which average to $6/6 = 1$.

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Two effects cancel almost perfectly: with more people, each is less likely to get their own item — but there are more people who could. That is why the answer barely depends on n.

Part B: The gambler's ruin

how likely are you to double your money?

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Most people guess "a bit under a half". Keep your guess; the answer comes later in these slides.

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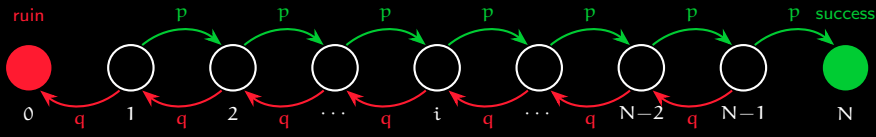
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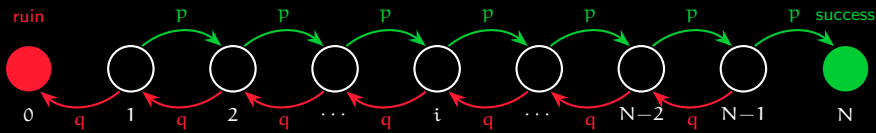
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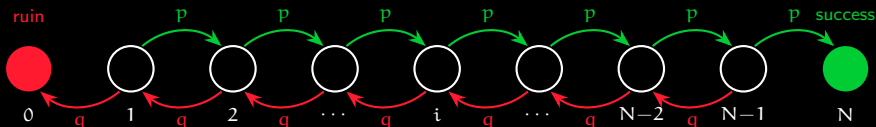
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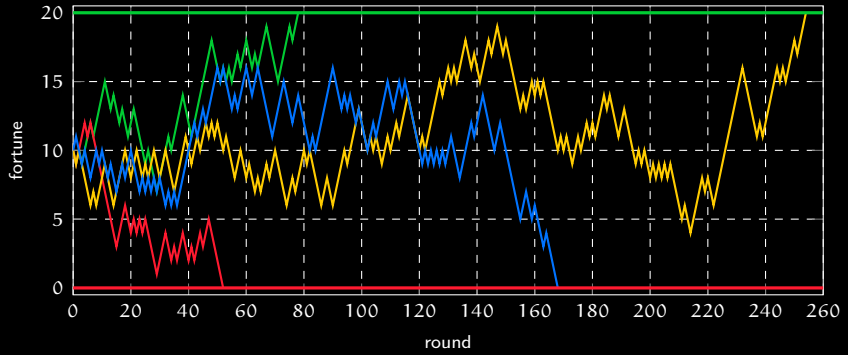
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Let P_i be the probability that the gambler reaches N before 0, starting from i. We want a formula for P_i.

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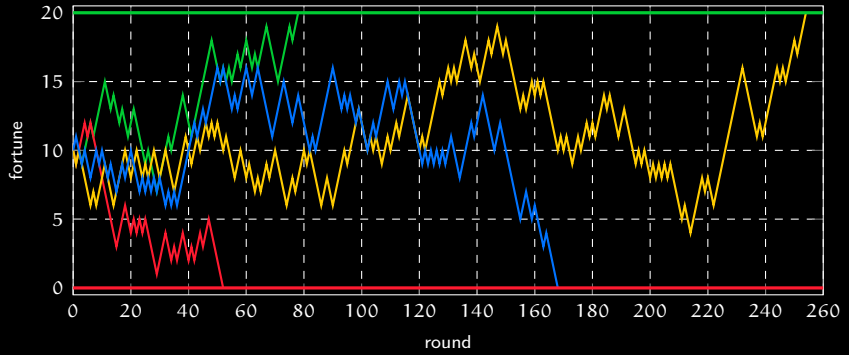
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Four runs of a fair game ($p = \frac{1}{2}$) starting from $i = 10$ with target $N = 20$:



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Four runs of a fair game ($p = \frac{1}{2}$) starting from $i = 10$ with target $N = 20$:



Two runs reach 20, two are ruined, and the games last anywhere from about 50 to over 250 rounds. Each run is a random walk stopped at the two boundaries.

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Theorem (gambler's ruin). Let $r = q/p$. Starting from i, the probability of reaching N before 0 is

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A special case of this problem appeared in Huygens' book on games of chance (1657); the general solution is due to James Bernoulli, published in 1713.

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So there are really only two outcomes, success and ruin. (Without this step, the two probabilities we are about to compute might not add up to 1.)

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This move — condition on the first step, and recognise the original problem inside itself — is called first-step analysis. The boundary values say that at 0 you have already lost and at N you have already won.

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So the increments $\Delta_i = P_i - P_{i-1}$ satisfy $\Delta_{i+1} = r\,\Delta_i$, with $r = q/p$. Starting from $\Delta_1 = P_1 - P_0 = P_1$:

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Note $r = 1$ exactly when $p = q = \frac{1}{2}$.

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Sanity checks: $P_0 = 0$, $P_N = 1$; and as $p \to \frac{1}{2}$, $r \to 1$ and $\frac{1-r^i}{1-r^N} \to \frac{i}{N}$ (divide top and bottom by $1 - r$), so the two formulas join up continuously.

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(expand: $px^2 - px - qx + q = px^2 - x + q$), so $x = 1$ or $x = q/p = r$.

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The same method solves every recurrence of this shape, including the one for the length of the game later in these slides.

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Let Q_i be the probability that the opponent ends up with everything. Swapping the roles of the two players swaps $p \leftrightarrow q$ and $i \leftrightarrow N - i$, so for $p \neq \frac{1}{2}$

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$$P_i + Q_i = \frac{p^N - q^N}{p^N - q^N} = 1.$$

(For $p = \frac{1}{2}$: $\frac{i}{N} + \frac{N-i}{N} = 1$.)

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Two independent routes agree: someone always wins. This is Ross's way of proving Step 1 — the block argument earlier in these slides gets there directly.

The fair game: $\mathrm{P_i} = \mathrm{i}/\mathrm{N}$

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- You have ₹10 and the casino has ₹10,00,000. Even at a perfectly fair game, you break the bank with probability $10/10{,}00{,}010 \approx 0.00001$.

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A quick calculation that becomes meaningful later in the course: your average final fortune is

$$N \cdot \frac{i}{N} + 0 \cdot \left(1 - \frac{i}{N}\right) = i,$$

exactly what you started with — whatever target N you choose. Setting a target to "quit while ahead" does not change your average in a fair game.

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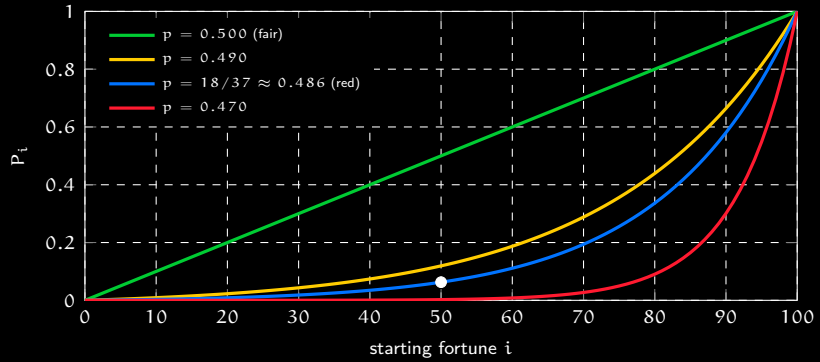
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In a fair game the richer player is not more skilful, just harder to ruin. The game transfers money, on average, nowhere — but it ruins the poorer player most of the time.

A small edge becomes a large one

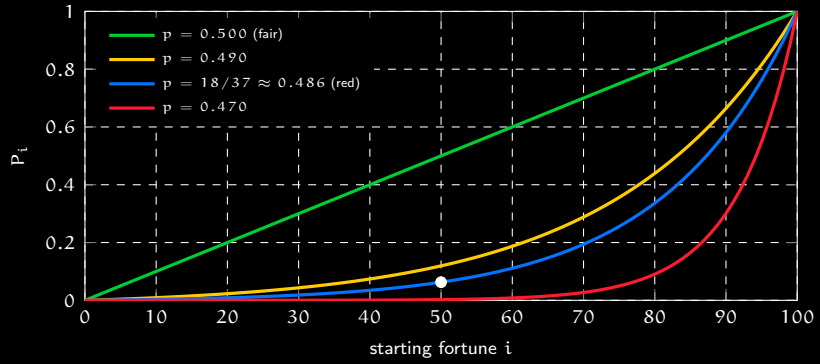
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Probability of reaching $N = 100$ before 0, as a function of the starting fortune i:



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Probability of reaching $N = 100$ before 0, as a function of the starting fortune i:



A disadvantage of barely 1.4 percentage points per spin (the blue curve) drags Ravi's chance of doubling ₹50 down to about 6.3% (white dot). The house edge is tiny per bet and enormous over a long game.

If you must gamble, gamble boldly

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Ravi still wants to turn ₹50 into ₹100 on red. What if he bets bigger stakes? With stakes of s rupees the problem is the same one with $\mathrm{i} = 50/s$ and $\mathrm{N} = 100/s$.

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₹2	25	50	0.206

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In an unfavourable game, play as few rounds as possible: bold play beats timid play. (The classic study is Dubins and Savage, *How to Gamble If You Must*, 1965.) In a favourable game, the advice reverses.

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What this limit means. Let R_N be "ruined before reaching N". Reaching $N+1$ requires passing through N, so $R_N \subseteq R_{N+1}$; and $\bigcup_N R_N$ is "ruined eventually". For an increasing sequence of events, $P\left(\bigcup_N R_N\right) = \lim_N P(R_N)$ (a consequence of Axiom 3; Ross, §2.6). Hence

$$P(\text{ever ruined}) = \lim_{N\to\infty} (1 - P_i) = \begin{cases} (q/p)^i & p > \frac{1}{2}, \\ 1 & p \le \frac{1}{2}. \end{cases}$$

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Against unlimited money, a fair or unfavourable game ruins you with certainty. With an edge ($p = 0.6$, $i = 10$) you survive for ever with probability $1 - (2/3)^{10} \approx 0.98$.

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Ross (§3.4) credits the question to Fermat; the dice version is Huygens' own. Note $1 - r^{24} = (1 - r^{12})(1 + r^{12})$, which gives the simplification above.

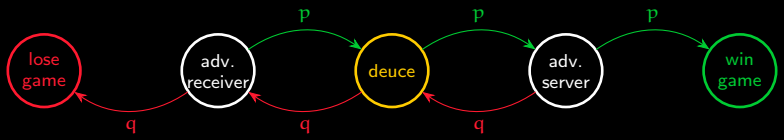
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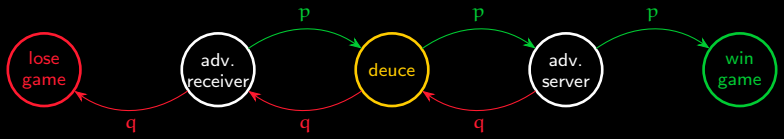
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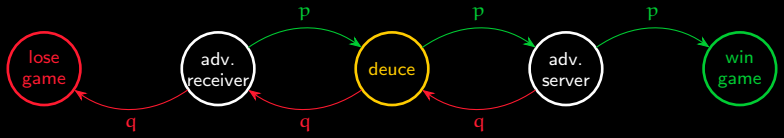


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$$P(\text{server wins from deuce}) = \frac{1 - r^2}{1 - r^4} = \frac{1}{1 + r^2} = \frac{p^2}{p^2 + q^2}.$$

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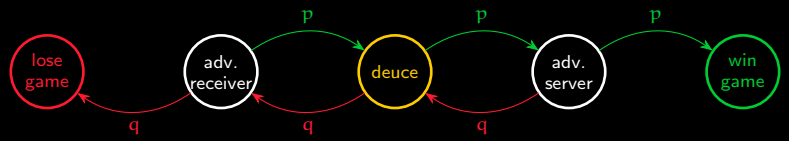
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A 55% edge on single points becomes a 60% edge from deuce. "Win by two" rules amplify small differences in skill — the same effect as the house edge, working for the stronger player.

Bonus: how long does the game last? (1/3)

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$$m_i = 1 + p\, m_{i+1} + q\, m_{i-1} \quad \text{for } 1 \le i \le N-1,$$
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Uniqueness. If m and m' both solve this system, their difference $d_i = m_i - m'_i$ satisfies $d_i = p\, d_{i+1} + q\, d_{i-1}$ with $d_0 = d_N = 0$. That is the equation for P_i with boundary values 0 and 0, and the argument of Step 3 shows its only solution is $d \equiv 0$ (all increments are multiples of d_1, and $d_N = 0$ forces $d_1 = 0$). So it is enough to find one solution.

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Proof. Boundary values: $m_0 = 0$ and $m_N = 0$. And for $1 \le i \le N-1$, since $(i+1)(N-i-1) + (i-1)(N-i+1) = 2i(N-i) - 2$,

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- Starting with just ₹1 against a friend with ₹99: on average $1 \times 99 = 99$ tosses — even though you are ruined 99% of the time, the rare lucky runs are very long.

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Ravi betting ₹1 on red from ₹50 towards ₹100 plays on average about 1618 spins — roughly 27 hours at one spin a minute — and still goes home broke 94% of the time. Betting all ₹50 at once: one spin.

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Whenever a derivation feels slippery, simulate. Ten lines of code will not prove a theorem, but they will catch almost any mistake in one.

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The same idea runs through both parts: condition on the first thing that happens, and look for a smaller copy of the original problem inside.

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with answers at the end

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7. Using the formula for P_i, show that when $p < \frac{1}{2}$ and $i = N/2$, $P_i < \frac{1}{2}$; that is, an unfavourable game remains unfavourable however far you are from either end.

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7 With $i = N/2$, $P_i = \dfrac{1 - r^i}{1 - r^{2i}} = \dfrac{1}{1 + r^i}$, which is $< \frac{1}{2}$ exactly when $r^i > 1$, i.e. $r > 1$, i.e. $p < \frac{1}{2}$.

Where to read more

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- **Ross**, *A First Course in Probability*: Example 5m of Chapter 2 (the matching problem by inclusion–exclusion), Example 2g of Chapter 3 (exactly k matches), Example 4l of Chapter 3 (the gambler's ruin, with Huygens' problem).

Where to read more

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- **Dubins and Savage**, *How to Gamble If You Must* (1965): the theory of bold play.



Questions?

manjil.saikia@ahduni.edu.in